

YUGESHKARAN P

GenAI Engineer · GenAI & Agentic Systems · LangChain · CrewAI · RAG · PyTorch
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PROFESSIONAL SUMMARY

GenAI Engineer with hands on experience in building and deploying production-grade agentic AI systems using Python, LangChain, CrewAI, and PyTorch. Hands-on expertise architecting multi-agent LLM workflows, RAG pipelines, and secure API backends with measurable production outcomes. Experienced in bridging AI prototypes to enterprise-grade deployments — implementing guardrails, rate limiting, JWT/OAuth authentication, Redis caching, and CORS security at the system level. Adept at co-building with technical teams and converting real-world AI integration challenges into scalable reusable solutions.

TECHNICAL SKILLS

Agentic AI & LLMs: LangChain, LangGraph, LangSmith, CrewAI, ReAct, multi-agent orchestration, MCP, tool-calling, Gemini
ML & Deep Learning: PyTorch, HuggingFace Transformers, NLP, RAG pipelines, embedding models, LLM inference optimization
Languages: Python, JavaScript (ES6+), TypeScript, SQL, C — data structures, algorithms, software design
Backend & Security: Flask, Node.js, Express.js — REST APIs, JWT/OAuth, rate limiting, CORS, Redis caching, guardrails
Cloud & DevOps: AWS (S3, Lambda, Amplify), Docker, Git/GitHub, CI/CD, Linux/Unix, Vercel
Databases & Vectors: MySQL, MongoDB, Redis, Pinecone VectorDB — semantic retrieval, schema design, query optimization

PROJECTS

Assistant Knowledge Hub — Production RAG System [Github](#)

Python · Flask · OpenAI · Pinecone VectorDB · LangChain · Pydantic

- Built a production-grade RAG assistant with OpenAI text-embedding-3-small and Pinecone VectorDB, enforced **20 requests/minute/user rate limiting** to optimize LLM API costs and ensure equitable resource allocation across concurrent users.
- Implemented **CORS middleware blocking 100% of cross-site request forgery** attempts and deployed **LLM guardrails preventing 100% of prompt injection attacks** — delivering enterprise-grade security for a production agentic deployment.
- Designed scalable semantic ingestion and retrieval pipelines with Pydantic schema validation and AGUI dynamic content rendering, architected for zero-downtime production operation under concurrent query load.

Mentor Consulting Crew AI — Multi-Agent System [Github](#)

Python · CrewAI · Gemini 2.0 Flash · ScrapeWebsiteTool · SerperDevTool

- Architected a production multi-agent system using CrewAI with hierarchical delegation and ReAct patterns, agents autonomously perform web research, scrape content sources, and generate structured curriculum plans with efficient task distribution across specialized roles.
- Integrated Gemini 2.0 Flash as the core LLM backbone with ScrapeWebsiteTool and SerperDevTool for live data retrieval, designed agentic workflows with self-reflection loops to improve output accuracy and reduce hallucination in research synthesis tasks.

Mini Language Model — Custom NLP Training [Github](#)

Python · PyTorch · Transformers · Flask

- Designed and trained a 2.96M-parameter transformer-based LLM from scratch using PyTorch.
- Built an end-to-end pipeline covering tokenization, architecture design, optimization, and inference speed and memory efficiency tuning and exposed via Flask inference API.

AI Assisted Tech Community Platform [Github](#)

Node.js · Flask · React.js · MongoDB · Redis · AWS S3 · Socket.io · JWT

- Implemented **Redis server-side caching** on content feed API achieving **80% reduction in API calls** per user session, replaced session-based auth with **JWT authentication** across all client requests, eliminating session-store performance bottlenecks.
- Engineered API-level **rate limiting middleware** preventing **100% of abusive requests and DoS attack vectors**, built real-time Socket.io discussions, AWS S3 media storage, and deployed on Vercel with CI/CD practices.
- Built responsive React.js/TailwindCSS frontend and integrated AWS S3 for scalable media storage, deployed on Vercel with CI/CD practices and maintained high availability under concurrent user load.

INTERNSHIP EXPERIENCE

Full-Stack Web Development Intern · Quantum Sharq Innovative Solutions

2024

- Built and optimized React.js/TailwindCSS web applications for clients, developed modular HRMS components in an Agile team, maintaining code quality through reviews and testing.

Data Science Intern · CompWallah IT Industry

2023

- Performed data analysis, ML model training, and Python diagnostic scripting on business datasets, built visualizations to drive data-informed decision-making workflows.

EDUCATION

B.Tech — Artificial Intelligence & Data Science · Dhanalakshmi Srinivasan University, Trichy

2021 – 2025

CGPA: **9.21 / 10.0** · Class XII (CBSE): 92%

RESEARCH PUBLICATIONS

- (Jun 2025)** "Evaluating the Performance of LLMs on SQL and NoSQL Databases Using LangSmith" — Intl. Conf. Multidisciplinary Solutions for Emerging Trends 2025
- (Sep 2024)** "Design and Development of an Attendance Management and Analysis System Using LLM" — ICIAAIEA 2024

Certifications: Python for Data Science · Data Analysis with Python · SQL & Relational Databases 101 · Data Visualization using Python